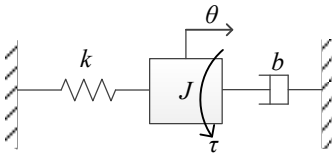
Huazhong University of Science and Technology
School of Mechanical Science and Engineering
System Dynamics
Homework 1

1. Consider a mass-spring-damper system as shown in Fig. 1. Derive the equation of motion. For the following inputs and initial conditions, take Laplace Transform and find Θ(s); derive the rotational displacement response θ(t) with Inverse Laplace Transform::



a) $\tau =$ unit step, $\theta(0) = \dot{\theta}(0) = 0$, $J = 1$, $b = 3$, $k = 2$.

b) $\tau = 0, \theta(0) = 1, \dot{\theta}(0) = 0$, $J = 1$, $b = 3$, $k = 2$.

Hint: For $\Theta(s) = L[\theta(t)]$, the Laplace Transform of its derivatives can be obtained as

$$L[\ddot{\theta}(t)] = s^2\Theta(s) - s\theta(0) - \theta'(0), \quad L[\dot{\theta}(t)] = s\Theta(s) - \theta(0)$$

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2. Linearize $y = \sin(\theta) + \cos(\theta) + \theta\cos(\theta) + \dot{\theta}\sin(\theta)$ around the equilibrium $\theta = \dot{\theta} = 0$

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3. Derive the dynamic equation of the 1/4 Car Model and find the response of $y_1(t)$ for a step change in y.

